Mycorrhizae as pollution indicators

Mycorrhizal fungi are not only good for the health of plants—they can also act as indicators of the health of the entire environment. Laboratory experiments with these fungi have shown that some grow badly in the presence of toxins, which means that they can be used to detect pollutants in the air or soil. For instance, some fungi fail to grow when exposed to heavy metals such as lead or cadmium, and because different kinds of fungi react differently to environmental change, certain species can be used to identify specific kinds of pollution.

Mycorrhizae are also useful indicators of the health of their native habitat. Many form cauliflower-like growths on tree roots, but these are smaller in polluted soil. The trees themselves may also respond to pollution with weaker shoot growth, but the mycorrhizal response is more acute and serves as a valuable early-warning sign of a habitat in decline.